MASSACHUSETTS INSTITUTE OF TECHNOLOGY
Department of Mathematics

Course Syllabus: 18.06 Linear Algebra

Course Description:
Basic introduction to linear algebra including systems of equations, vector spaces, matrices, determinants, eigenvalues, and positive definite matrices.

UNIT 1: Vectors and Linear Combinations
- Vector operations: addition and scalar multiplication in 2D and 3D space
- Linear combinations of vectors and spans
- The dot product, vector lengths, unit vectors, and angles
- Matrix-vector products and representations of linear equations

UNIT 2: Solving Linear Equations and Matrix Algebra
- Gaussian elimination and row operations
- Triangular matrices and back substitution
- Matrix multiplication rules and inverses
- LU Decomposition and factorization (A = LU)
- Permutation matrices and row exchanges (PA = LU)

18.06 Linear Algebra Syllabus (Continued)

UNIT 3: Vector Spaces and Subspaces
- Vector spaces and fundamental subspaces of a matrix
- The column space and null space of A
- Solving Ax = 0 and the pivot variables
- The complete solution to Ax = b
- Linear independence, basis, and dimension
- The four fundamental subspaces and the Fundamental Theorem of Linear Algebra

UNIT 4: Eigenvalues and Eigenvectors
- The characteristic equation: det(A - lambda I) = 0
- Calculating eigenvalues and eigenvectors of 2x2 and 3x3 matrices
- Diagonalizing a matrix (A = S Lambda S inverse)
- Applications to difference equations and matrix powers
- Symmetric matrices and the Spectral Theorem